Analysis of electromagnetic waves on a macroscopic scale using microwaves

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The radiation dispersion of a Gunn-Diode as a microwave-transmitter was recorded and the full width at half maximum (FWHM) angle was found to be $(14.4 \pm 1.2)^\circ$. With this diode the focal length of a paraffin wax lens was determined to be $f = (39.18 \pm 0.39)$ cm. In addition, a polystyrene cube was examined using Bragg diffraction and the lattice constant of a supposed crystal inside was found to be $d_{\text{avg}} = (4.16 \pm 0.52)$ cm. Furthermore, frustrated total reflection was observed and the polarisation of the emitted microwave was investigated. Finally, one observed the propagation of microwaves in a metallic wave guide.

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1 Introduction

The importance of microwaves in modern society is significant. Used in communication systems or e.g. microwave-ovens one encounters microwaves in everyday life. Due to their longer wavelength they can also be easily used to demonstrate wave-mechanics on a macroscopic level. In the following experiments a Gunn-diode is used as a microwave-transmitter. For this diode the wavelength of the emitted radiation is measured using standing waves as well as a Michelson interferometer. Also, one observes the radiation dispersion with a high-frequency diode. In the second part of the experiment microwaves are used to demonstrate electromagnetic wave mechanics on a macroscopic length scale; e.g. the focal length of a paraffin wax lens is measured and the lattice constant of a polystyrene cube with a supposed crystal consisting of metallic objects arranged on a lattice in it is determined. In addition, one demonstrates quantum tunneling with two paraffin prisms. The polarisation of the microwaves is checked via a metallic grid and their propagation in a waveguide is observed.

2 Experiment

As source for the microwaves a Gunn-Diode is used. Attached to the transmitter was a metallic, rectangular horn. A similar horn was attached to the high-frequency diode that was used as a receiver in all other experiments except of subsection 2.3.

2.1 Radiation dispersion of a Gunn-Diode

Using a Goniometer the characteristic radiation behaviour of the Gunn-Diode was recorded. Therefore, the Diode is centered on the Goniometer and a photo diode is used as a detector. In Figure 1 the normalized intensity is shown in dependency of the recorded angle. As can be seen in Figure 1 , the dispersion is nearly symmetric around 90°. A downward drift of the measured current was observed. This was probably caused by the discharging battery that is used to drive the receiver. Since the current values for the angles from 0° to 88° were taken after the values from 90° to 180°, they were affected most by the drift. This is suggested to be the main cause

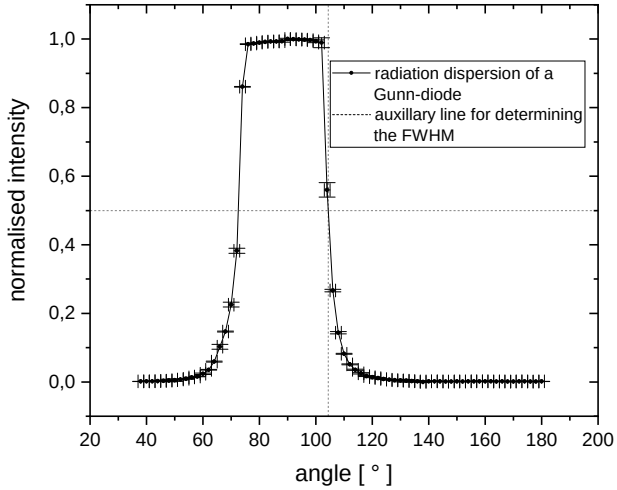

Fig. 1: The normalised intensity of the used Gunn-Diode depending on the angle. Here an angle of 90° means that the receiver was directly in front of the transmitter.

for the asymmetry observed in Figure 1. The break in the peak at around 14 degrees (104° in Figure 1) is the most apparent evidence for that; note the visible one percent drop in the intensity after the measurement of the right side, wich took about seven minutes.
Because of the better reliability, the right flank was used for determining the full width at half maximum (FWHM) angle. The guidelines indicate it's position. The FWHM angle is found to be $(14.4 \pm 1.2)°$. One conjectures this might be attributed to the shape of the horn which had an opening angle of a similar magnitude.

2.2 Focal length of a paraffin wax lens

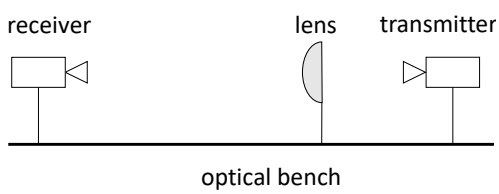

Fig. 2: Experimental set up for determining the focal length.

The Gunn-Diode was placed on an optical bench. In a distance of about 80 cm a paraffin wax lens was also placed on the optical bench. The receiver was placed at about 90 cm away from the lens and then gradually moved towards it while the receiver-voltage was recorded. The setup for this part is depicted in Figure 4. Due to reflections on the metallic horn and the mounting of the wax lens, standing waves overlapped the signal of the focused microwaves. In order to measure correct amplitudes only the peaks of the standing waves were measured. The data is shown in Figure 4. Since the detected voltage is depending on the intensity of the incoming microwave radiation the position of the projected image can be found by searching for the maximum in current in Figure 4. According to Newton's imaging equation for a thin lens, meaning a lens which radius is small compared to its thickness,

$$\frac{1}{g} + \frac{1}{b} = \frac{1}{f} \tag{1}$$

where g is the distance from transmitter to lens, b is the distance from lens to image and f is the wanted focal length. Given Figure 4 the position of the image was estimated to be at $b = (73 \pm 1)$cm. For this estimation an exponential fit of the form $e^{a+cx+dx^2}$ was performed considering all data except of the red ones. With Equation 1 the focal length can be calculated straightforward as $f = (38.17 \pm 0.32)$cm. The relatively low error can be attributed to the high reading accuracy of the length measurements. However, due to the mounting of both receiver

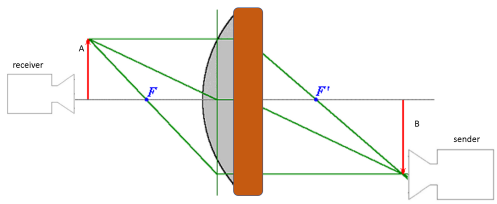

Fig. 3: Difficulties in setting up the receiver according to the optical path. Even though one tried to adjust the distance to the optical axis of sender and receiver, this could not be realized sufficiently. This resulted in an unwanted distance of the receiver and sender to the optical axis. The optimal distances are denoted as A and B.

and transmitter an error was caused by missing the correct optical path. In Figure 3 this is depicted. The figure shows the error in adjusting the height of the receiver. It was probably not set correctly according to the height B of the transmitter. Hence, the image could have been projected above or below the receiver, shifting the position of the maximal intensity. Moreover, one was not able to access the inner parts of the lens and estimate their error. Thus, a larger error stands to reason but could not be exactly quantified due to a lack of insight into the direct structure of the lens.

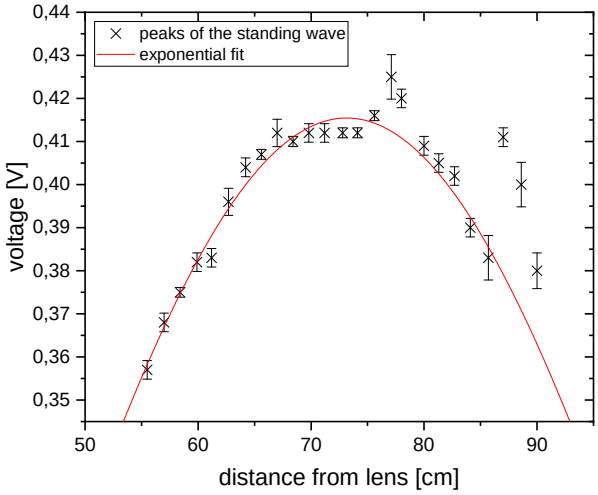

Fig. 4: Measurement of the peaks of the standing waves for determining the focal length of the paraffin wax lens. To estimate the point of maximal intensity an exponential fit of the form $e^{a+cx+dx^2}$ was performed. The red points were probably caused by interference of reflecting microwaves on experimenter and environment and are therefore not considered by the fit.

2.3 Determination of the wavelength via standing waves

The wax lens was removed and instead a metal plate was placed on the optical bench in a distance of about 1 m and used as a reflector to create standing waves. As a receiver a Diode was used. With this setup the number of wave nodes between the transmitter and the metal plate were counted. Together with the path length of those 50 nodes, the wavelength is calculated via the relation

$$\lambda = \frac{2l}{z} \tag{2}$$

where l is the measured length and z is the number of nodes counted. With the measurement of 50 nodes Equation 2 results in a wavelength of $\lambda = (2.876 \pm 0.024)$cm. The error is caused by the error of the length measurements. On the label of the microwave-transmitter the frequency was given as 10.5 GHz which corresponds to a wavelength of about 2.855 cm. The two results are compatible within the range of accuracy. For the speed of light $c = 299\,792\,458\,\frac{\mathrm{m}}{\mathrm{s}}$ [1] was used.

2.4 Determination of the wavelength via the Michelson interferometer

Alternatively to the measurement above the wavelength was deduced using a Michelson interferometer. Via a difference in the optical path length (OPL) the well known interference experiment was realized. While varying the distance of the second reflector one is able to observe constructive and destructive interference. Counting the maxima encountered while moving the second reflector, one is able to calculate the wavelength as follows;

$$\lambda = \frac{2d_m}{z}. \tag{3}$$

where z is the number of maxima that were detected while the mirror was moved the distance d_m. Limited by the experimental setup only 20 nodes including the starting node could be measured. The error was determined by the read off accuracy of the covered path length. The wavelength was found to be $\lambda = (2.853 \pm 0.042)$cm. This value is compatible with the manufacturer specification. This result coincides with the one obtained in subsection 2.3 within the limits of accuracy. The relative error is nearly twice as big. Conversely, in subsection 2.3 more than twice as many nodes were considered and therefore the measurement was much more precise.

2.5 Total reflection and frustrated total reflection

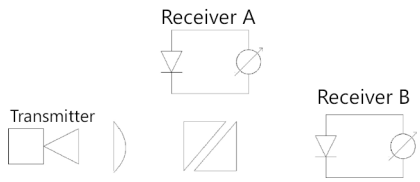

Fig. 5: The setup to observe total reflection. The microwave-transmitter is aimed at the two prisms. With the two receivers A and B the reflected and transmitted intensity is measured.

Considering a boundary between two different isotropic media with refractive index n_1 and n_2, the incident angle α of an incoming electromagnetic wave and β of the outgoing wave are according to Snell's law related by

$$\frac{\sin\alpha}{\sin\beta} = \frac{n_2}{n_1} \tag{4}$$

. If $n_1 > n_2$ then there is an angle α_t for that $\beta = 90°$. For $\alpha \geq \alpha_g$ total reflection occurs. Nevertheless an evanescent wave still propagates along the surface with an amplitude that decays exponentially with distance from the interface.

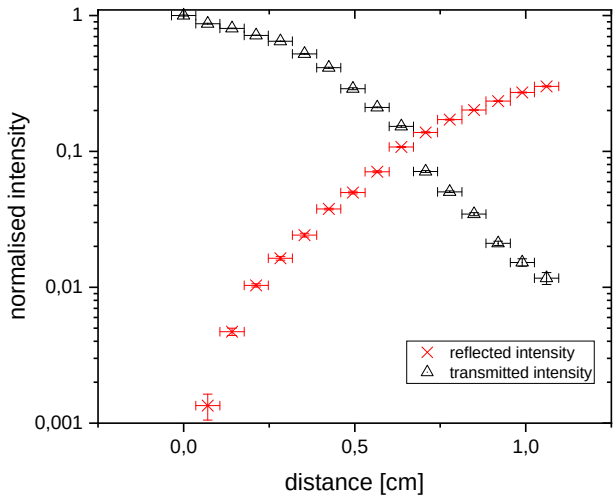

Fig. 6: The normalized measured transmitted and reflected intensity. As one grows the other shrinks. At a distance of 0 cm the whole intensity is transmitted and not reflected.

The intensity follows

$$I_{\text{transmitted}} \propto \mathrm{e}^{-\frac{2z}{s}} \tag{5}$$

with

$$s = \frac{\lambda \cos\alpha}{2\pi\sqrt{n_1^2 \sin\alpha^2 - n_2^2}}. \tag{6}$$

This evanescent wave can be captured if a second medium with similar refractive index is brought close to the surface where the total reflection takes place. The totally reflected wave can then be detected in the second medium even if the two media do not touch. This is called frustrated total reflection. This is analogous to the effect of quantum tunneling where the amplitude of a particle wave doesn't vanish instantly at a potential barrier but decreases exponentially. With the Gunn-Diode and two paraffin prisms the effect was observed using an assembly shown in Figure 5. The maximal intensity that was detected by the receiver B was used to normalize all other intensities. The signal of the receiver A at a distance of $d = 0\,\mathrm{m}$ was set to zero to correct all other intensities. The results are shown in Figure 6. The first data point of the reflected intensity is 0 and therefore not shown in the logarithmic graphic. It can be seen that the transmitted intensity decreases with increasing distance of the prisms while the reflected intensity increases.

2.6 Polarisation of the used microwaves

If the horn of the receiver is rotated 90° relatively to the horn (shown by position 2 in Figure 7) of

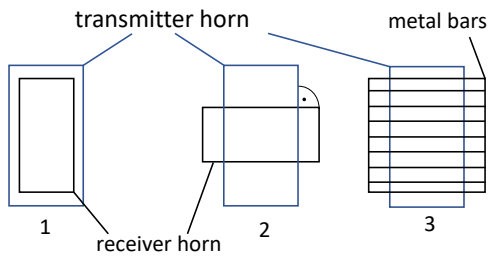

Fig. 7: Depicted are the outlines of the horns of receiver and transmitter as they would be seen from the position of the Gunn-Diode. In position 2 no radiation was detected. This suggests that the emitted microwaves are linear polarised. In position 3 with metal bars between transmitter and receiver no signal was detected.

the transmitter no signal was detected. This suggests that the microwaves are linearly polarised. This idea is also supported by the following experiment. Metallic bars were put between receiver and transmitter. The horns of the receiver and the transmitter had the same orientation (shown by position 1 in Figure 7). The detected voltage was recorded while the bars were rotated. The results are shown in Figure 8.

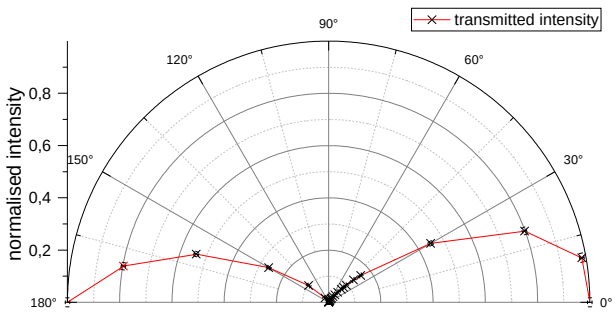

Fig. 8: The intensity going through the bars and reaching the receiver. At 0 degree and 180 degree the intensity peaks. The bars were vertically oriented. At 90° the intensity vanished The whole radiation is blocked by the bars.

The measurement could have been affected by a slight tilt of the receiver and/or the Gunn-diode since their mounting was neither flat nor stable. If the bars are oriented perpendicular to the long side of the horn on the Gunn-diode the whole intensity is blocked. This constellation is shown as position 3 in Figure 7. The microwaves with an amplitude parallel to the metallic bars are reflected and therefore not transmitted. The metallic bars act as a filter for the microwave radiation. Since it is dependent on the angle of the bars it has to be a polarization filter. The radiation completely vanishes at 90° degrees, meaning that all incoming waves are reflected. Furthermore this

means that the emitted microwaves have only an amplitude along this direction. This leads to the conclusion that the used microwaves have to be linear polarized parallel to the short side of the horn of the transmitter.

2.7 Determination of lattice constant of a cubic crystal

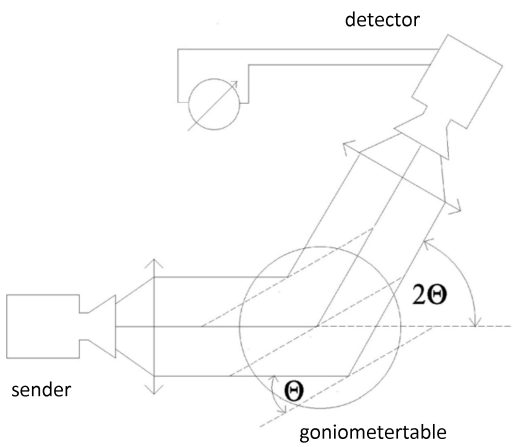

Fig. 9: The sender gives microwaves of length 2.8cm on the cubic crystal. With each measurement latter is turned by the angle Θ while the receiver is turned by the angle $2 * \Theta$. By this procedure one realizes the symmetric wave incidence on which the Bragg formula is conditioned.

If the wavelength of the used light is known it can be used to analyse a crystal via Bragg-diffraction. The following relation holds:

$$n\lambda = 2d \sin\theta \tag{7}$$

where n is the order of the observed reflex, λ the wavelength of the used radiation, d the distance between the crystal planes and θ the angle under which the maximum is detected. This shows that with a given wavelength λ the Bragg condition can only be applied for $\lambda \leq 2d$. With the wavelength of 2.855 cm (see subsection 2.3) crystals down to $d \approx 1.438$ cm can be analysed. The setup for this experiment is shown in Figure 9 The above formula wouldn't take the orientation of the observed crystal plane with Miller index h, k, l into account. The formula is corrected with the knowledge that for a simple cubed crystal

$$d_{hkl} = \frac{d}{\sqrt{h^2 + k^2 + l^2}} \tag{8}$$

where d_{hkl} is the distance between similar planes with Miller index h, k, l. The correct formula then reads

$$d = \frac{n\lambda}{2\sin\theta}\sqrt{h^2 + k^2 + l^2}. \tag{9}$$

The cube was scanned from two different sides; first in the (100) plane and after that in the (110) plane. These are depicted in Figure 10. The side

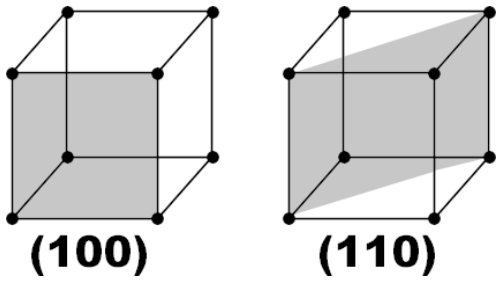

Fig. 10: Crystal planes for a cubic crystal with their corresponding Miller index below. [3]

of the cube that was examined was placed parallel to the microwave beam path in the 0° position. The data obtained for the intensity measurement of the (100) plane dependent on the orientation is shown in Figure 11. The peak at an angle of 0° is not caused by the crystal but by the straight beam from transmitter to receiver. Therefore, it is irrelevant for the evaluation.

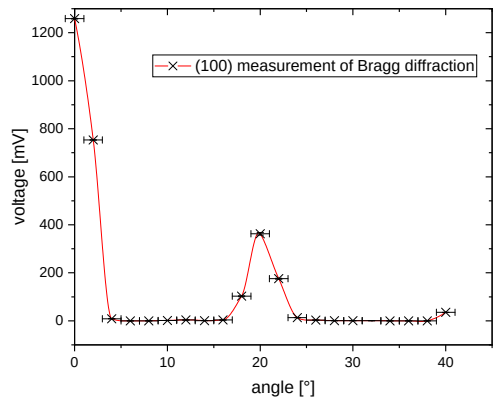

Fig. 11: The detected voltage in a Bragg diffraction experiment for a (100) of a cubic crystal; The first order diffraction maximum is located at $\approx 20°$. Also, one is able to see the first bit of the second order maximum at 40°.

After receiving these interference pictures similar to ones of crystals on the atomic length scale one assumes the polystyrene cube to contain a metallic grid or other electromagnetic wave bending material. Thus, the Bragg formalism that was introduced earlier in the section can be applied; Assuming a regular metallic structure in

the polystyrene cube, one determines the lattice constants from the different sides as follows:

$$d_{(100)} = (4.25 \pm 0.28)\text{cm} \tag{10}$$

$$d_{(110)} = (4.06 \pm 0.52)\text{cm} \tag{11}$$

$$d_{\text{avg}} = (4.16 \pm 0.52)\text{cm} \tag{12}$$

As one can see both the lattice constant are compatible with each other within the range of their accuracy. Note that the different errors result from the different relative errors that were used for the error propagation calculation. Considering the overall size of the observed cube this lattice constant seems plausible. As in the case with crystals and light on the atomic length scale, one can see how interference phenomena of electromagnetic waves can be used to deduce material characteristic data.

2.8 Waveguide

One tests a metallic hose's property as a waveguide, meaning a structure guiding electromagnetic waves with a minimal loss of energy [2]. In this manner one places the Gunn - Diode at one end and on the other end the receiver. By applying a closing cap on the one end one can verify that the electromagnetic wave travels in the hose instead of on its surface; If the closing cap is applied and the electromagnetic wave cannot be detected, one can conclude that the electromagnetic wave travelled in the hose, assuming that it was detected when the cap was removed. Also, the receiver diode was turned in the xy-plane by 180°. After doing so one noticed that there was no signal perceived anymore and thus one was able to conclude that the Gunn-Diode is emitting linear polarized light.

3 Summary

In this experiment we used the wave property of electromagnetic radiation on the microwave length scale to determine their polarization, dispersion behaviour in air as well as wavelength in varying set ups. First, one found the angular dependence of microwave radiation by measuring the intensity of the radiation from different angles. We found the radiation of the Gunn Diode to be distributed around a peak at 0°. After that one used the Michelson Interferometer and standing waves to determine the wave length of the Gunn-Diode with $\lambda = (2.853 \pm 0.042)$cm and $\lambda = (2.876 \pm 0.024)$cm. Then, one determined the focal length of a paraffin wax lens and one was able to see how microwaves share the refraction properties of light in the visible spectrum. In this manner one also looked at phenomena like total and frustrated total reflection. One ascertained the linear polarization of the microwaves by attaining the intensity of the microwaves after passing through metal bars at different angles. As it is possible to determine the lattice constant of a crystal with the Bragg condition on the atomic length scale, in the present case one deduced -on a macroscopic length scale- the lattice constant of a polystyrene cube with a probably metallic lattice structure in it. The experiment was finalized by demonstrating that a metallic hose can indeed be used as a waveguide.

References

[1] `https://physics.nist.gov/cgi-bin/cuu/Value?c`

[2] `https://en.wikipedia.org/wiki/Waveguide`, last visited September 5th

[3] `https://de.wikipedia.org/wiki/Datei:Miller_Indizes_Ebenen.png` (cropped), last visited September 5th

[4] reference of figure 9 :

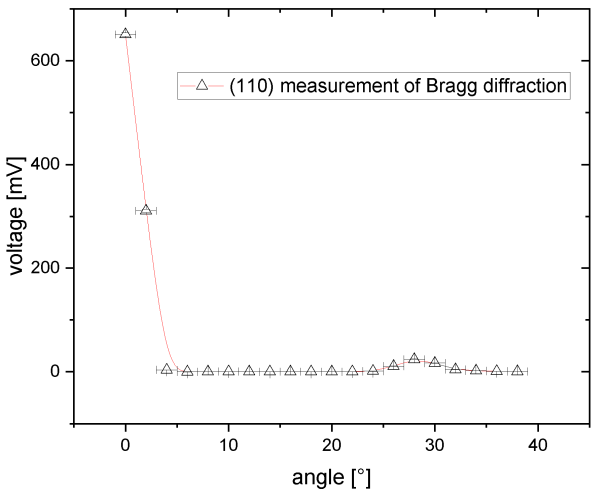

Fig. 12: The diffraction of the (110) plane explaining the procedure of the analysis of the (110) and (100) plane experimental data. For estimating the position of the maximum a B - Spline in Origin 2020 was used to fit the experimental data (picture). The first and only maximum is located at ≈ 28°